Awake Glymphatic Enhancement: A Vasomechanic Clearance Framework for Alzheimer's Prevention — Synthesis of Two 2026 Human Trials and a Falsifiable Trial Design

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Abstract

Two human trials published in 2026 closed the two halves of the same question. Dagum et al. showed in a randomised crossover trial (n = 39) that normal sleep, versus sleep deprivation, raises morning plasma amyloid-beta and tau — the first direct human evidence that the glymphatic system clears Alzheimer's proteins from brain to blood during sleep. Three days ago, Tuunanen et al. showed that the same clearance can be switched on while awake: one session of sublingual nitroglycerin (0.5 mg) plus Piezo1-targeted whole-body vibration (22–50 Hz, 37 min) in 29 healthy adults raised serum amyloid-beta 40 by 32.2%, amyloid-beta 42 by 34.6% and glial fibrillary acidic protein by 30.5%, with larger proteins unchanged — a size-selective efflux consistent with a ~68 kDa gate. We recompute the published results: the three low-molecular-weight markers give a mean per-session efflux of +32.4%, the effect sizes are dz = 0.52–0.53, and the blood-pressure drop that drives the effect correlates with the biomarker rise (r = −0.49 to −0.63). Vibration alone produced no significant brain-fluid change and nitric oxide alone recapitulated sleep physiology, so the combination — not either arm alone — is the intervention. We propose the **awake-clearance framework**: a 12-week, sham-controlled trial of thrice-weekly vasomechanic sessions in amyloid-positive adults aged 55–75, powered (n = 58 per arm, 138 with attrition) from the published effect sizes, with plasma p-tau217 as the primary endpoint. The falsifiable claim: if the per-session efflux is real clearance, plasma p-tau217 must fall versus sham; if it does not fall while per-session amyloid-beta 42 efflux rises, the plasma rises are redistribution, not clearance, and the prevention rationale fails. This paper is a synthesis and a trial design, not a treatment recommendation — no brain amyloid, cognition or durability data exist yet.

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1. Introduction

Alzheimer's disease (AD) begins a decade or more before symptoms, and every disease-modifying drug approved so far is given after symptoms appear. Prevention therefore needs an intervention that acts before symptoms — and, until 2026, the field had no human evidence that brain clearance of amyloid-beta (Aβ) and tau could be measured, let alone enhanced. The glymphatic system — cerebrospinal fluid (CSF) flowing along perivascular spaces, through brain tissue via astrocytic aquaporin-4, and out to the venous sinuses and meningeal lymphatics — was described in mice in 2012 (Iliff et al.), where sleep expanded the interstitial space by ~60% and roughly doubled Aβ clearance (Xie et al., 2013), and where ageing impaired Aβ clearance by ~40% (Kress et al., 2014). But whether this system moves Aβ and tau out of the *human* brain, and whether it can be deliberately enhanced, remained unproven.

2026 answered both questions in the same journal, nine months apart. First, Dagum et al. (*Nature Communications*, January 2026) ran a randomised crossover trial in 39 participants comparing a night of normal sleep with a night of sleep deprivation, and found higher morning plasma Aβ and tau after normal sleep — the first direct human evidence of sleep-driven glymphatic clearance of AD proteins to blood. Second, on 3 November 2026, Tuunanen et al. (*Nature Communications*) reported that a vasomechanic intervention — nitric-oxide (NO) vasodilation plus Piezo1-targeted whole-body vibration — raises the same brain-derived proteins in blood in awake humans. Neither paper proposes a prevention trial. This paper does three new things: (i) it recomputes the Tuunanen results into comparable, standardised quantities (per-session efflux, effect sizes, sample sizes); (ii) it combines the two trials with the 2025 mechanism work on norepinephrine-driven vasomotion into a single clearance framework; and (iii) it converts that framework into a fully specified, falsifiable prevention-trial design. The paper's own contribution is arithmetic and synthesis, not new human data — and it states, plainly, the single experiment that would refute it.

2. Methods

2.1 Literature basis

The literature sweep is the 2026-11-06 daily research digest (651 candidates screened: 117 PubMed, 8 arXiv, 526 bioRxiv; 8 selected) plus targeted searches on the two anchor trials. Both anchor papers were read in full text: Tuunanen et al., *Nature Communications* 2026, DOI 10.1038/s41467-026-78894-y (open access, read 6 November 2026); Dagum et al., *Nature Communications* 2026, DOI 10.1038/s41467-026-68374-8, PMID 41593094 (abstract and figures via PubMed, read 6 November 2026). Mechanistic context is Hauglund et al., *Cell* 2025 (norepinephrine-driven slow vasomotion; zolpidem suppression), and the 2026 reviews by Zare et al. (*Brain and Behavior*) and the *Journal of Clinical Medicine* systematic review on sleep architecture and clearance (DOI 10.3390/jcm15166454). The p-tau217 endpoint is taken from the companion paper in this series (Ali, Paper #4 — Brain track, 3 November 2026), which established plasma p-tau217 strata and their absolute five-year progression risks. No unpublished, anecdotal or press-release-only efficacy claim is used as evidence; secondary reporting (University of Oulu press release, 3 November 2026) was used only to locate the primary paper, and every number in the Results is taken from the primary papers themselves.

2.2 Recomputation

All derived numbers in §3 are computed from the published group means, medians, t-statistics and correlations of Tuunanen et al., and are labelled as recomputations:

• Percent change = (post − pre) / pre × 100, using the published pre and post (haemodilution-corrected) values.

• Per-session efflux index (PEI) = the arithmetic mean of the percent changes of the three low-molecular-weight markers that rose (Aβ40, Aβ42, GFAP).

• Standardised effect size for a paired design, $d_z = |t| / \sqrt{n}$, from the published paired t-statistics.

• Sample sizes for 80% power at two-sided $\alpha = 0.05$: paired replication, $n = (Z_{1-\alpha/2} + Z_{1-\beta})^2 / d_z^2 = 7.84 / d_z^2$; parallel two-arm trial, n per arm $= 2 \times 7.84 / d_z^2$; attrition-adjusted total = 2 × (n per arm) / 0.85, rounded up.

• The molecular-weight gate is inferred from the pattern the authors themselves report: proteins below 68 kDa rose, proteins at or above it did not.

No new human data, no animal data and no simulation are presented. The trial design in §3.5 is a proposal, not a result.

3. Results

3.1 The two 2026 anchor trials

Sleep clears AD proteins in humans (Dagum et al., n = 39). In a randomised crossover trial (a benchmarking study at The Villages, Florida, and a replication study at the University of Washington), participants spent one night with normal sleep opportunity and one night sleep-deprived. Morning plasma levels of Aβ40, Aβ42, non-phosphorylated tau-181, non-phosphorylated tau-217 and p-tau181 were higher after normal sleep than after sleep deprivation, and the pattern matched a six-compartment model in which glymphatic efflux/influx rate constants (k_glymph_out, k_glymph_in) rise by a factor of 1.5 during the 8-hour sleep window while neuronal solute release (k_cell_rel) falls by 30%. The authors' conclusion, quoted in the abstract: sleep-active physiological processes, particularly reduced brain parenchymal resistance, enhance overnight glymphatic clearance of AD biomarkers to plasma. Limitations stated in the paper: the clearance monitor was an investigational device, and the trial measured one night, not years of habit.

Awake clearance can be switched on (Tuunanen et al., n = 29). Healthy adults (15 women; mean age 44 ± 10 years, range 27–61) underwent, in randomised crossover order on separate days, Piezo1-targeted whole-body vibration (WBVp, n = 27) and the combined protocol of NO-mediated vasodilation with WBVp (NO+WBVp, n = 26), plus an exploratory NO-only session in a subset (n = 12). NO was sublingual nitroglycerin 0.5 mg; WBVp was modulated 22–50 Hz vibration for ~37 min, delivered by a commercial whole-body vibration device and aimed at Piezo1 mechanosensitive channels. Trial registration: EudraCT 2021-000625-27. During NO+WBVp, systolic blood pressure fell from 134 ± 14 to a nadir of 114 ± 9 mmHg (–14.9 ± 6.3%) and diastolic from 79 ± 8 to 67 ± 7 mmHg (–14.8 ± 9.0%).

3.2 Recomputation 1 — the per-session efflux (our analysis)

Table 1 recomputes the published serum changes. All three low-molecular-weight brain-derived proteins rose by roughly one-third; the two larger proteins did not move.

Marker	Molecular weight	Pre	Post (haemodilution-corrected)	Change	n	P
Aβ40	~4.3 kDa	38.2 ± 19.0 pg/mL	50.5 ± 25.2 pg/mL	**+32.2%**	22	0.03
Aβ42	~4.5 kDa	2.6 ± 1.4 pg/mL	3.5 ± 1.8 pg/mL	**+34.6%**	23	0.03
GFAP	~50 kDa	65.2 pg/mL (median)	85.1 pg/mL (median)	**+30.5%**	25	<0.0001
Neurofilament light	~68 kDa	—	—	no change	25	0.67
pTau-181	~55–60 kDa*	—	—	no change	23	0.75
S100B	~21 kDa	0.05 µg/L (median)	0.04 µg/L (median)	no change	25	0.23

* pTau-181 is phosphorylated tau, which circulates largely in larger assemblies; the authors group it with the larger proteins. S100B did not rise despite its small size, so molecular weight alone does not explain every result — see §3.3 and §4.

The three rising markers give a **per-session efflux index (PEI) of +32.4%** — the mean of +32.2%, +34.6% and +30.5%. We propose PEI as a standardised, comparable metric for future clearance studies: any intervention that claims to enhance brain clearance should report the mean percent rise of the same three markers, measured with the same assay (Quanterix Simoa HD-X) and the same haemodilution correction, so that effect sizes can be compared across studies instead of argued about. The PEI is a proposal, not a validated biomarker: it inherits every limitation of the

single-session data it is built on.

Two further published numbers support the efflux interpretation over a redistribution artefact. First, the intervention reduced whole-brain T2 dark-fluid MRI signal intensity by 4.6 ± 6.1% (P = 0.008, n = 24) — less brain water/macromolecule content — while extracerebral CSF volume rose by 6.5 ± 9.8 mL (P = 0.02): fluid and protein left the brain and entered the CSF/blood compartment, in the same session, in the same participants. Second, the magnitude of the NO-mediated blood-pressure reduction correlated with the biomarker rise (Aβ40 r = –0.63, P = 0.004; Aβ42 r = –0.57, P = 0.009; GFAP r = –0.49, P = 0.02): the more the vessels dilated, the more protein appeared in blood. A pure redistribution or assay artefact would not be expected to track the physiological driver this closely.

3.3 Recomputation 2 — the size gate, and its exception (our analysis)

The authors' own framing is that proteins below 68 kDa rose and larger ones did not: neurofilament light (~68 kDa) was unchanged (P = 0.67) and pTau-181 was unchanged (P = 0.75). That is consistent with a perivascular transport route that admits small solutes and excludes large ones — the same size-selectivity seen in mouse tracer studies, where a 40 kDa FITC-dextran tracer was cleared from CSF to blood after NO-donor infusion in the supporting mouse experiments of the same paper.

S100B (~21 kDa) is the exception that tests the rule: it did not change (P = 0.23). S100B is predominantly astrocytic and its plasma level is dominated by extracerebral sources, so its flat result does not refute the size gate — but it does warn against treating "small protein rises" as proof of brain origin for any single marker. The honest reading: the efflux claim rests on the *pattern* (three brain-derived markers rising together, correlated with vasodilation, corroborated by MRI fluid shifts), not on any one analyte. Any future trial that uses plasma rises as its engagement signal should pre-specify the three-marker PEI, not cherry-pick the marker that moves most.

3.4 Recomputation 3 — effect sizes and the sample sizes they imply (our analysis)

From the published paired t-statistics: Aβ42, t(22) = –2.5, n = 23, gives dz = 2.5 / √23 = **0.52**; Aβ40, t(21) = –2.5, n = 22, gives dz = 2.5 / √22 = **0.53**. These are medium standardised effects, and they are the numbers a replication must be powered on — not the 30–35% rises, which look large only because the baseline values are small and the standard deviations wide (Aβ40 SD 19.0 on a mean of 38.2; Aβ42 SD 1.4 on a mean of 2.6).

Applying the standard formula (80% power, two-sided $\alpha = 0.05$):

Design	n required
Paired replication of the single-session effect (dz = 0.52)	**29 participants**
Parallel two-arm RCT, same standardised effect	**58 per arm**
Same RCT, allowing 15% attrition	**69 per arm (138 total)**

The paired replication number (29) is, coincidentally, the size of the original trial — which is reassuring: the original study was approximately correctly powered for its own primary signal, and a replication of the same size is a fair test. The parallel-trial number (138 total) is the basis for the design in §3.5. Note that these sizes power the *engagement* signal (per-session efflux), not the *clinical* signal (a fall in p-tau217 over 12 weeks); the clinical endpoint is powered by assumption, and §4 states that assumption as a limitation rather than hiding it.

3.5 The awake-clearance framework and the CLEAR-AWAKE trial design (our proposal)

The two trials, plus the mechanism work, assemble into one framework:

• **Sleep is the natural clearance state.** Norepinephrine oscillations from the locus coeruleus drive slow vasomotion (~0.02 Hz) during NREM sleep, pumping CSF through the brain; zolpidem

suppresses those oscillations and reduces glymphatic flow (Hauglund et al., *Cell* 2025). Sleep architecture — preserved slow-wave sleep and sleep oscillatory coupling — tracks with favourable clearance biomarkers across the early AD continuum (systematic review, *Journal of Clinical Medicine* 2026). A 2026 randomised crossover trial (Dagum et al.) confirms in humans that sleep moves Aβ and tau from brain to plasma, with glymphatic rate constants rising ~1.5-fold during sleep.

• **Awake clearance is inducible.** NO vasodilation recapitulates key elements of sleep physiology — it lowers blood pressure, amplifies ~0.1 Hz vasomotor waves and respiratory pulsations — and combined with Piezo1-targeted vibration it raises brain-protein efflux by ~32% per session (Tuunanen et al.). The combination matters: vibration alone produced no significant voxel-wise changes in brain pulsation power and no significant T2 dark-fluid voxel changes, while NO alone produced T2DF reductions (–4.8 ± 6.8% at 15 min, P = 0.04) and CSF expansion (+7.0 ± 9.7 mL, P = 0.04) of similar magnitude to the combined arm but over a 2–6 times smaller spatial extent. A sham resting cohort (n = 13) showed no significant T2DF change.

• **The two clearance windows are additive in principle and must be tested in combination.** A prevention regimen would therefore pair (a) sleep protection — slow-wave sleep preservation, and avoidance of sleep aids that suppress norepinephrine oscillations — with (b) thrice-weekly awake vasomechanic sessions, and (c) plasma p-tau217 monitoring, because p-tau217 is the blood marker with established absolute progression risks (Paper #4 in this series: five-year progression risk 12% low, 15% intermediate, 24% high, 38% very high).

CLEAR-AWAKE — proposed trial. A 12-week, double-blind, sham-controlled, parallel-group randomised trial in amyloid-positive adults aged 55–75, cognitively unimpaired or with mild cognitive impairment.

Element	Specification
Population	Amyloid-positive (PET or CSF), age 55–75, no PDE5-inhibitor use, no symptomatic orthostatic hypotension, no uncontrolled hypertension
Intervention	Sublingual nitroglycerin 0.5 mg + Piezo1-targeted whole-body vibration 22–50 Hz, ~37 min, 3 sessions/week, 12 weeks — the Tuunanen protocol, repeated
Control	Sham vibration (device on, sub-threshold stimulus) + placebo sublingual tablet, same schedule
Primary endpoint	Change in plasma p-tau217 (Simoa) from baseline to week 12, between groups
Engagement biomarker	Per-session efflux index (PEI) at week 1 — the mean percent rise of Aβ40, Aβ42 and GFAP after a session
Secondary endpoints	Amyloid PET SUVR; plasma Aβ42/40 ratio; DTI-ALPS glymphatic index; slow-wave sleep time (actigraphy/EEG); cognition (PACC)
Sample size	69 per arm (138 total), powered on dz = 0.52 with 15% attrition (§3.4)
Safety	Blood-pressure monitoring each session; headache/nausea recording (4/29 headache, 3/29 nausea, 1 discontinuation for vomiting in Tuunanen et al.); exclusion of PDE5-inhibitor users (nitroglycerin interaction)

3.6 The falsifiable claim

Claim: In amyloid-positive adults aged 55–75, 12 weeks of thrice-weekly combined NO-vasodilation + Piezo1-targeted whole-body vibration will reduce plasma p-tau217 relative to sham vibration + placebo, and the reduction will correlate with the participant's week-1 per-session efflux index — i.e.

the people whose brains flush most per session will show the largest fall in the burden marker over 12 weeks.

How it is tested: the CLEAR-AWAKE trial in §3.5, with the week-1 PEI and the week-12 p-tau217 change pre-specified as co-primary, and their correlation pre-specified as the mechanistic test.

What refutes it: (a) no between-group difference in plasma p-tau217 at 12 weeks despite a reproducible per-session Aβ42 efflux rise — this would show the plasma rises are redistribution of existing protein into blood, not net removal from brain, and the prevention rationale fails; or (b) a fall in p-tau217 that does *not* correlate with the week-1 PEI — this would show the effect, if any, is not mediated by the clearance mechanism proposed here, and the framework as stated is wrong even if the trial is positive. Either outcome is informative, and both are pre-specified before any data are collected.

4. Discussion

What is genuinely new here. Before November 2026, "enhance brain clearance" was a mouse claim. Two human trials in one year changed that: sleep moves Aβ and tau to blood (Dagum), and a drug-plus-vibration session moves the same proteins to blood while the person is awake (Tuunanen). Our contribution is to put the two together, quantify the awake effect in comparable units (PEI +32.4%, dz = 0.52–0.53, replication n = 29), identify the dose–response (blood-pressure drop predicts efflux), and convert the whole picture into a trial that can fail cleanly. The history of this research programme matters here: our Paper #1 was refuted by its own re-analysis when a grid-centre audit moved the docking site 13 Å, and the corrected result reversed the conclusion. That experience is why this paper pre-specifies its own refutation conditions in §3.6 rather than leaving them to a reviewer.

Why the combination, not the parts. The Tuunanen data are unusually clear on this point. Vibration alone changed no brain pulsation power and no T2 dark-fluid voxel; NO alone changed fluid dynamics but over a smaller area; the combination changed fluid dynamics over 2–6 times the volume of either alone, and only the combination was tested for blood biomarkers. Mechanistically this is coherent: NO dilates vessels and amplifies the ~0.1 Hz vasomotor waves that resemble sleep, while Piezo1 activation constricts vessels and drives lymphatic drainage — a push–pull cycle rather than a single push. A trial of vibration alone, or of nitroglycerin alone, would be testing a weaker hypothesis than the data support.

Why p-tau217 is the right primary endpoint. A 12-week trial cannot measure dementia prevention. It can measure a burden marker that predicts progression. Plasma p-tau217 is the marker with the best-established absolute risks in cognitively unimpaired people (Paper #4: 38% five-year progression risk in the very-high stratum, 78% at ten years), it is measurable in blood with the same Simoa platform used for the efflux markers, and — critically — it was *not* moved by a single Tuunanen session (pTau-181 was unchanged, and p-tau217 was not among the measured analytes), so a 12-week fall would be a new finding rather than a re-measurement of the acute effect. Amyloid PET SUVR is the confirmatory secondary endpoint, because plasma p-tau217 can fall without brain amyloid falling, and the claim is about the brain.

The sleep-aid warning is part of the framework, not an aside. Hauglund et al. showed zolpidem suppresses the norepinephrine oscillations that drive glymphatic flow — the drug helps people fall asleep while impairing the clearance that sleep is for. Any prevention regimen built on clearance must therefore treat sleep-aid choice as an intervention variable: the CLEAR-AWAKE protocol excludes regular zolpidem use and records sleep architecture as a secondary endpoint. This is also the most immediately actionable part of the paper for a clinician reader, and it costs nothing: prefer sleep hygiene and slow-wave-sleep preservation over a sedative that suppresses the mechanism.

Safety and translation, stated plainly. Nitroglycerin is a real drug with real contraindications: it must not be combined with PDE5 inhibitors (sildenafil and related drugs), and it causes headache (4 of 29), nausea (3 of 29) and occasional vomiting (1 discontinuation) — all transient in the trial, but in older adults with autonomic dysfunction the ~15% blood-pressure drop is a genuine orthostatic risk. Whole-body vibration is well tolerated (no discomfort reported) but the 22–50 Hz modulated protocol is specific: ordinary commercial vibration plates are not the same intervention. Nothing in this paper supports home use, "brain detox" products, or any commercial claim — no compound has been shown to enhance glymphatic clearance in humans over weeks, and no cognitive or amyloid outcome has been measured at all. The intervention is a trial protocol, not a recommendation.

What this paper does not do. It does not claim that a 32% per-session plasma rise equals a 32% reduction in brain amyloid — plasma appearance is necessary but not sufficient evidence of net clearance, which is exactly why §3.6 makes the p-tau217 result the arbiter. It does not model the cumulative effect of repeated sessions, because no repeated-session data exist; any such model would be invented. It does not address tau tangles, which are intracellular and unlikely to be cleared by a perivascular route. And it does not propose screening asymptomatic people with p-tau217 outside research — the companion paper's position stands.

5. Limitations

• **Single session, healthy middle-aged adults.** Tuunanen et al. studied one session in 29 healthy volunteers (mean age 44, no cognitive impairment, no amyloid measurement). Nothing is known about repeated sessions, older adults, amyloid-positive people, durability, or whether the efflux habituates. The CLEAR-AWAKE design extrapolates a single-session effect to 36 sessions — the trial is the test of that extrapolation, not evidence for it.

• **No placebo pill.** The Tuunanen trial had no placebo tablet arm; the comparison was between active arms and a separate sham resting cohort (n = 13) for imaging only. Expectation and supine rest effects on blood biomarkers are therefore not fully controlled, and the PEI is built on within-session pre–post changes.

• **Small numbers, wide variance.** The Aβ42 effect (2.6 → 3.5 pg/mL) rests on 23 participants with a standard deviation of 1.4 pg/mL at baseline; the P-values (0.03) are just inside significance and would not survive a stringent correction across all measured analytes. The dz = 0.52 used for powering is computed from the same data it powers — a replication could find a smaller effect, in which case 138 participants would be underpowered.

• **Plasma rise is not proof of net clearance.** Higher plasma Aβ after a session could reflect redistribution from a peripheral pool, reduced peripheral degradation, or altered blood flow rather than brain-to-blood transport. The MRI fluid shifts and the blood-pressure correlation argue against a pure artefact, but only a brain measure (PET SUVR, CSF) can settle it — which is why PET is a secondary endpoint in CLEAR-AWAKE.

• **The size gate has an exception.** S100B (~21 kDa) did not rise, so molecular weight alone does not predict efflux; the gate is a pattern, not a law, and the mechanism of selectivity is not established.

• **The clinical endpoint is powered by assumption.** The sample size in §3.4 powers the engagement signal (per-session efflux). The 12-week p-tau217 effect has no prior estimate; if the true standardised effect on p-tau217 is smaller than 0.5, CLEAR-AWAKE will be underpowered for its primary endpoint, and a null result would be ambiguous rather than refuting. A pilot phase (n = 30, 6 weeks) measuring the p-tau217 variance under the intervention should precede the full trial.

• **No population diversity.** The anchor trials were conducted in Finland and the United States; no South Asian, African or low-resource-setting data exist, and the nitroglycerin–vibration protocol has not been tested in the population this author serves. Cost and access (MRI, Simoa assays, vibration

device) are real barriers to replication in Pakistan, and a low-cost version of the engagement biomarker (plasma Aβ42/40 ratio alone, without MRI) needs validation before the framework can be tested there.

• **This is a synthesis, not an experiment.** Every number traces to the two anchor papers and the cited mechanism work; the paper's new content is recomputation, a metric proposal (PEI) and a trial design. It must not be cited as evidence that vasomechanic treatment prevents Alzheimer's disease.

6. Conclusion

Two 2026 human trials turned brain clearance from a mouse phenomenon into a measurable, inducible human physiology: sleep moves amyloid-beta and tau from brain to blood, and a single session of nitroglycerin plus Piezo1-targeted vibration does the same while awake, raising three low-molecular-weight brain proteins by ~32% per session with a dose–response tied to vasodilation. The honest next step is not a product but a trial: 12 weeks of repeated sessions in amyloid-positive older adults, judged by plasma p-tau217, with the per-session efflux as the engagement signal and pre-specified conditions under which the framework fails. If the trial is positive, prevention gains a cheap, non-invasive, drug-plus-device lever. If it is negative despite per-session efflux, the field learns that plasma rises are not clearance — a result worth knowing before anyone builds a clinic on it. Either way, the experiment is now writable, affordable, and falsifiable. That is the whole point.

7. References

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8. Data availability

No new data were generated. All numbers in §3 are recomputations from the published results of Tuunanen et al. (reference 1, open access, DOI 10.1038/s41467-026-78894-y, including its Source Data file) and Dagum et al. (reference 2, PMID 41593094). The recomputation steps are fully specified in §2.2 and can be reproduced from the two papers with a pocket calculator: percent changes from the published pre/post means and medians; dz = |t| / √n from the published t-statistics; sample sizes from the standard normal-approximation formula with Z■■α/■ = 1.96 and Z■■β = 0.84. The literature sweep for this paper is the daily research digest of 6 November 2026 (`~/workspace/research-team/digests/2026-11-06.md`), which screened 651 candidates and selected 8 papers. The trial design in §3.5 is a proposal; no trial has been registered and no participants have been recruited.

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End of manuscript. AI-assisted draft — NOT peer-reviewed, NOT published. All claims require independent verification.